Exam Date & Time: 22-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

VII Sem - End Semester Examination - Nov-Dec 2024

DRUG DELIVERY [BME4063]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions. Section Duration: 180 mins

1A) Analyse and predict the feasibility of the type of passive diffusion possible for each drug based on the following data: (3)

Drug	Log P	Log $D_{6.0}$
Atenolol	0.10	-2.74
Famotidine	-0.40	-2.06
Ibuprofen	3.72	2.12

1B) Differentiate transcellular and paracellular transport. Explain how an uptake transporter plays the role of efflux transporter in a carrier mediated drug distribution process. (4)

1C) Pindolol has a volume of distribution (V_d) of 2300L. If the plasma concentration is 1mg/L, (3)

(i) How much of drug is in the body?

(ii) How much of drug is in the plasma? (assume that the volume of plasma is 3L)

(iii)How much of drug is in the tissue?

2A) Illustrate graphically and mathematically, the influence of tissue and plasma protein binding on the pattern of drug distribution. (4)

2B) Deduce an expression for distribution when drug binds neither to plasma protein nor tissue protein (i.e f_u=1 and f_{ut}=1) and interpret the impact on volume of distribution. (3)

2C)	Justify with logic how the following factors influence renal tubular reabsorption (i) the drug's lipophilicity, (ii) pH and (iii) filtrate flow rate.	(3)
3A)	Consider an extraction unit, where in Ca=160mg/L, Cv=100 mg/L, and Q=2L/h. Find out the rate of extraction, clearance and the fraction extracted.	(4)
3B)	Highlight the advantages of transdermal drug delivery system.	(3)
3C)	Explain in detail the different classes of 'acquired immunity'.	(3)
4A)	After collecting of blood (from toxoid treated horse), how would you proceed to obtain dry product (powder) of diphtheria antitoxin?	(2)
4B)	Compare the methods of preparation of smallpox vaccine.	(5)
4C)	How do you assess whether an individual is susceptible or immune to diphtheria? Explain the process with logic.	(3)
5A)	Highlight the key functions of targeted drug delivery system.	(3)
5B)	Differentiate active and passive targeting.	(3)
5C)	Analyse the process pertaining to ultrasound mediated targeted drug delivery.	(4)

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